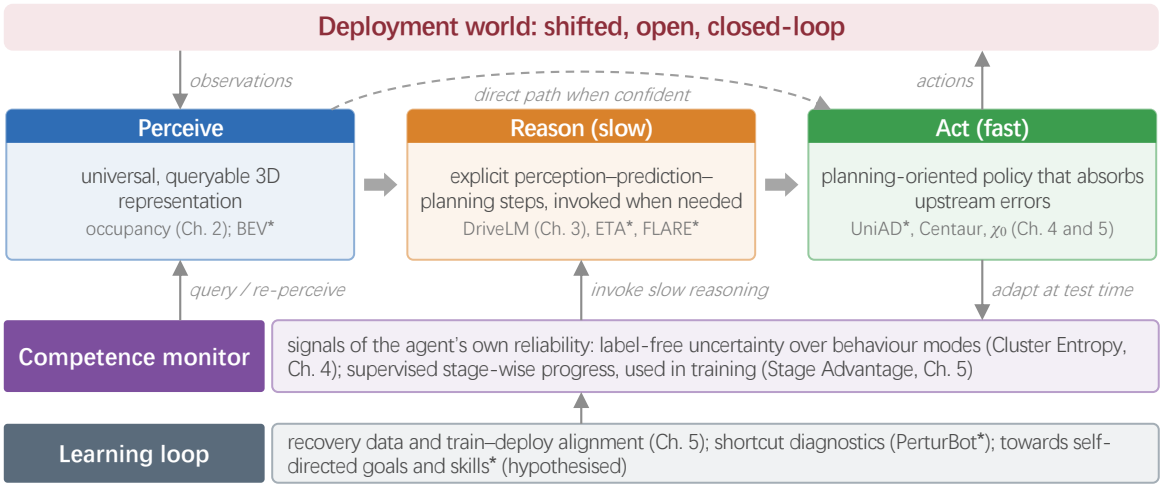

Deployment world: shifted, open, closed-loop
observations
direct path when confident
actions
Perceive
universal, queryable 3D representation
occupancy (Ch. 2); BEV*
Reason (slow)
explicit perception–prediction–planning steps, invoked when needed
DriveLM (Ch. 3), ETA*, FLARE*
Act (fast)
planning-oriented policy that absorbs upstream errors
UniAD*, Centaur, χ_0 (Ch. 4 and 5)
query / re-perceive
invoke slow reasoning
adapt at test time
Competence monitor
signals of the agent's own reliability: label-free uncertainty over behaviour modes (Cluster Entropy, Ch. 4); supervised stage-wise progress, used in training (Stage Advantage, Ch. 5)
Learning loop
recovery data and train–deploy alignment (Ch. 5); shortcut diagnostics (PerturBot*); towards self-directed goals and skills* (hypothesised)